SIDDHI PRAVIN LIPARE

MS by Research · CVIT, IIIT Hyderabad

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EDUCATION

MS by Research, Computer Science & Engineering — Jan 2026 – Present
International Institute of Information Technology, Hyderabad — CGPA 8.00

BS Data Science & Engineering — Dec 2021 – May 2025
Indian Institute of Science Education & Research, Bhopal — CGPA 8.41

Senior Secondary (Class XII), CBSE Board · 90.0% — 2021

Secondary (Class X), CBSE Board · 95.0% — 2019

EXPERIENCE

Centre for Visual Information & Technology (CVIT), IIIT Hyderabad — May 2025 – Dec 2025
Applied Research Fellow — Offer Letter

- Designed **RoadTones** – a dataset–model–evaluation framework for tone-controllable road-video captioning. Work accepted at **CVPR Findings 2026** (Project page).
- Developed key components of the dataset generation pipeline and evaluation metrics, enabling comprehensive benchmarking against 10 state-of-the-art Video-Language Models (VLMs).
- Investigated and optimized the vLLM inference pipeline for the project demo, minimizing video quality degradation during inference to preserve caption generation performance.

Indian Institute of Technology Madras — May 2024 – Jul. 2024
Research Intern — Certificate, Report

- Explored Super-Resolution techniques for 3D medical imaging, enhancing z-axis resolution in CT and MRI scans.
- Utilized GANs, Diffusion Models, and NeRF to improve image clarity for better diagnostics.

PROJECTS

BS Thesis – Image Tampering Detector — Jan. 2025 – April 2025
IISER Bhopal — Report

- Leveraged the use of Discrete Wavelet Transform along with an Adaptive Sub-Band Attention Module to capture frequency-spatial cues to enable precise localization of tampered regions in an image.

Improving Monocular Depth Estimation of Outdoor Images in Dim Light Conditions — Sept. 2024 – Nov. 2024
IISER Bhopal — Report

- Integrated denoising and deblurring filters into the transformer-based DepthAnything V2 model to improve depth estimation in blurry and noisy nighttime images.

Prediction of Mortality Rate of Heart Failure Patients Admitted to ICU — Sept. 2023 – Nov. 2023
IISER Bhopal — GitHub

- A binary classification problem executed using supervised machine-learning techniques. Several classifiers and pre-processing methods were applied to determine which classifier gave the best results.

TECHNICAL SKILLS

Languages: Python3, C/C++, JavaScript, MATLAB, R, HTML, CSS, MySQL

Technologies: PyTorch, TensorFlow, Keras, OpenCV

Other: Docker, Firebase, Android SDK, Data Structures & Algorithms, Linux

KEY COURSES

Mathematics: Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability & Statistics

DS & Engineering: 3D Deep Learning, Natural Language Processing, Data Analyses & Statistics for Geosciences, Applied & Accelerated AI, Deep Learning, Artificial Intelligence, Machine Learning, Computer Vision, Database Management Systems, Applied Optimisation, Econometrics, Data Science in Practice (Advanced Python), Data Structures & Algorithms, Signals & Systems

ACHIEVEMENTS

Publication — *'RoadTones: Tone Controllable Text Generation from Road Event Videos'* accepted at the CVPR 2026 Findings Track. — 2026

Presentation — Poster *'Improving Image Tampering Detection: A Hybrid Approach Using Frequency and Spatial Domain Features'* presented at CV4Science Workshop, CVPR 2025. — 2025

ISRO Robotics Challenge — Among the Top 30 teams out of 400 in India. — 2024

IEEE CAS Student Design Competition — 3rd place in the Asia-Pacific Region. — 2022